Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection

R26-IT-108

Project Proposal Report

Abeysinghe S.A.M.M.– IT22081148

B.Sc. (Hons) in Information Technology Specialization in Information Technology (IT)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

2026 March

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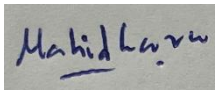
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this proposal is my own work and does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

Name	**Student ID**	**Signature**
Abeysinghe S.A.M.M.	IT22081148	

The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

Supervisor: Mr. Nelum Amarasena

Date: 12.03.2026

Co-Supervisor: Samadhi Rathnayake

Date: 12.03.2026

ABSTRACT

Cardiovascular diseases are among the most frequent causes of morbidity and mortality in the general population, and older adults are particularly prone to the disease because of physiological alterations and the greater risk of not having noticed any cardiac anomalies. Constant surveillance can be used to aid in early diagnosis and timely intervention, but most of the current systems are hospital-centric, costly wearable or cloud-based systems, which may jeopardize privacy, add latency, and are inaccessible in low-resource environments. This study offers the concept and implementation of a low-cost, wearable, and privacy-conscious cardiovascular anomaly detection system to monitor the elderly with photoplethysmography (PPG) sensing and Edge Artificial Intelligence. The suggested prototype combines the acquisition of physiological signals using the MAX30102 sensor with an ESP32 microcontroller to perform processing on the device, which provides the possibility to monitor in real-time without constant cloud connectivity. The study seeks to offer a practical solution applicable in daily life within the elderly health monitoring by centering on embedded and energy efficient architecture.

The study will be characterized by signal acquisition, preprocessing, features, model creation, and application via TinyML to ESP32. To detect abnormal cardiovascular patterns on processed physiological signals, a lightweight one-dimensional convolutional neural network (1D-CNN) will be developed. This will be followed by optimization of the trained model and conversion to embedded implementation with TensorFlow Lite to microcontrollers to achieve cost-effective memory and low inference times. The performance of the system will be measured based on accuracy of classification, response time and real-time applicability in a wearable setting. It is anticipated that this research will produce a usable prototype that can enable the detection of early cardiovascular anomalies at a low cost, portable, and without breaching the privacy of the end-user. The study advances research on the emerging domain of intelligent wearable healthcare by showing that Edge AI can support localized elderly monitoring applications.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CVD	Cardiovascular Disease
CNN	Convolutional Neural Network
ECG	Electrocardiogram
Edge AI	Edge Artificial Intelligence
ESP32	Espressif 32-bit Microcontroller
IoT	Internet of Things
MAX30102	Pulse Oximetry and Heart-Rate Sensor Module
PPG	Photoplethysmography
TinyML	Tiny Machine Learning

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1. INTRODUCTION

1.1 Overall Background of Research

1.1.1 Domain Overview

The ageing of the population has led to an increasing demand for a healthcare system that can support constant observation beyond the clinical care facilities. Understanding of the early warning signs is also delayed, and chronic illness, abrupt physiological decline, and chronic disease are more prone to older adults, who may live independently. In this respect, geriatric health care monitoring has come to refer to a regular periodical evaluation of health care in a hospital setting to an active and consistent evaluation with the help of digital health tools [3], [10].

The use of wearable health monitoring systems is of special significance since it allows non-invasive and real-time measurement of physiological signals in the context of daily life. Wearable devices can sustain long-term monitoring in home settings unlike traditional bedside monitoring, which imposes an additional burden on caregivers and healthcare facilities. What makes them practical is that they can provide essential data on a regular basis, identify abnormal physiological trends and provide early warning signals to preventive action [3], [10]. Due to the continued improvement of wearable sensing technologies, they are now being thought of as applicable to personalized, remote, and scalable healthcare delivery.

Cardiovascular monitoring is one of the most significant among numerous physiological functions, which can be observed in older adults. Cardiovascular anomalies like arrhythmias can be sporadic and symptomless, which implies that it would be challenging to identify without regular clinical visits only. In the case of older people, a late diagnosis of such abnormalities may result in serious complications such as stroke, heart failure and sudden cardiac events [1], [4], [5]. Thus, the concept of cardiovascular monitoring embedded in wearable elderly care systems is extremely relevant to aid in early detection and immediate medical intervention.

1.1.2 Background of the Problem

CVD is one of the leading health issues in older adults, and physiological ageing predisposes the patient to rhythm disruptions and other heart defects. Arrhythmias are particularly important as they can be unnoticeable or intermittent, and thus, they can be hard to detect without constant monitoring. In most instances, the symptoms may not be recognized immediately, or at all, by older adults, or they may not be evident until the condition has taken on a clinical severity [4], [5].

Unrecognized arrhythmias pose significant risk to the care of the elderly. Delay or missed diagnosis can be a contributor to complications of diminished cardiac efficiency, thromboembolic, hospitalization, and higher mortality rates. The common diagnostic methods like periodic recordings of electrocardiograms work well within a controlled environment but may not be feasible in everyday life. Consequently, wearable methods to aid regular or real-time cardiovascular evaluation outside the hospital are receiving increased interest [1], [3], [4].

Continuous real-time monitoring is especially necessary in the home setting where most of the ageing population resides without supervision or with minimal monitoring. More traditional methods of monitoring tend to be inadequate in such settings since they rely on regularly scheduled examination or symptom-induced clinical visits. This gap can be addressed by real-time wearable systems which allow continuous monitoring and quicker detection of abnormalities when they occur. This can be particularly helpful to older users, whose condition can decline very quickly without urgent outside attention [3], [5], [10].

1.1.3 Significance of Research

The importance of early detection of cardiovascular anomalies relates to the possibility of enhancing the likelihood of prompt clinical action and eliminating the possibility of turning a condition into a critical circumstance. Early identification of irregular heart rhythms can potentially minimize health risks, enhance timely medical decision-making, and enhance the overall safety of self-isolated elderly people [4], [5]. A monitoring system that can identify abnormal cardiovascular patterns in real time thus possesses not only clinical but also social value.

Live alarms have direct advantages to the elderly and their caregivers as well. To the user, instant detection could enhance the sense of personal safety and confidence in the period of independent living. In the case of caregivers and family members, early alerts will help minimize uncertainty and fast track responses in case of an emergency or even a suspected cardiac episode. Within the broader healthcare settings, these systems can also serve to minimize unnecessary hospital admissions through taking preventive action earlier [3], [10].

Simultaneously, new health monitoring solutions should also deal with privacy issues. Most of the existing smart monitoring systems use cloud-based analysis, where physiological data is sent to remote servers where they are processed. Although this strategy can be used to facilitate advanced analytics, it simultaneously creates issues surrounding data privacy, latency, network reliance, and security [6], [7]. Such issues are especially critical in the field of healthcare, where physiological information is especially sensitive. This is why privacy-preserving methods founded on Edge AI and TinyML are now more relevant since it allows local processing on embedded gadgets without the need to transmit data to the cloud continuously [6]– [8]. The study is thus not only important in that it focuses on detecting cardiovascular anomaly but also delves into a safer and more responsive method that is applicable in monitoring of elderly healthcare.

1.2 Background and Literature Survey

1.2.1 Relevant Studies and Techniques

Traditionally, cardiovascular monitoring has been based on the use of techniques that utilize electrocardiograms since the ECG signal is a direct measure of electrical activity of the heart. Basic studies and reference data sets like the MIT-BIH Arrhythmia Database have significantly contributed to arrhythmia detection algorithm design and test [1], [2], [12]. ECG is also very useful in clinical diagnosis, and numerous automated arrhythmia detection papers have relied on ECG recordings, owing to the diagnostic richness and standardized annotation frameworks [1], [11], [12].

Along with this, photoplethysmography has attracted interest as a wearable cardiovascular monitoring method. The appeal of PPG sensors is that they are inexpensive, non-invasive,

small and can be incorporated in wrist-worn or finger-based devices. It has been demonstrated in previous research that PPG-based monitoring can assist in estimating heart rate, as well as be investigated in detecting rhythm abnormalities, such as atrial fibrillation and similar irregularities [3], [4]. PGP is usually better matched to continuous consumer-grade wearable monitoring than ECG hardware, but the signal quality of motion and ambient conditions may interfere with PPG performance [3].

Signal preprocessing is thus an important component of biomedical sensing systems. Raw physiological signals are usually noisy, have baseline drift and motion artifact, which lower the reliability of detection. The signal quality requires enhancement through efficient preprocessing to improve the quality of classification or anomaly detection. A large body of literature on wearable physiological monitoring repeatedly focuses on filtering, artefact removal, segmentation and feature preparation as required steps to achieve meaningful model input [3], [4], [10].

Due to this ability to acquire discriminative patterns directly through physiological data, deep learning has gained importance in the field of arrhythmia detection. Convolutional neural networks and similar structures have been used to detect abnormalities in the heart with high accuracy on test sets [4], [11]. These techniques minimize the use of manual feature engineering and have demonstrated potential to represent delicate temporal and morphological variations relating to irregular heart rhythms. Deep learning hence provides a promising avenue toward enhanced performance in automated classification in the case of cardiovascular anomaly detection.

More recently, Edge AI and TinyML have become applicable in implementing machine learning on low-power embedded devices. Instead of uploading physiological information to the cloud to be processed remotely, these solutions enable local inference to run on microcontrollers. Embedded artificial intelligence systems based on TensorFlow Lite Micro and similar systems have demonstrated that neural network inference can be scaled to constrained platforms and thus is becoming more applicable to real-time medical tasks [6], [7]. The work on memory-efficient inference and model optimization has also contributed to the viability of embedded intelligence to wearable systems [8].

1.2.2 State of the Art

The present-day state of the art in heart monitoring involves integration of commercial wearable technologies, cloud-based monitoring systems, and research systems based on machine learning to detect anomalies. Smart wearable products are popular in monitoring heart rates and other physiological measures in everyday life. These systems reveal the feasibility of continuous health monitoring using wearable form factors in practice and have helped to make non-invasive digital health technologies more accepted by the public [3], [5].

At the research stage, cardiovascular anomaly detection has evolved to more intelligent and automated methods. Research on PPG-based arrhythmia detection demonstrates that optical sensing of the wearable heart rate can be combined with appropriate signal processing methods and machine learning to also detect abnormal rhythm patterns [4]. Moreover, benchmark studies based on existing cardiovascular datasets are still actively working on the design of anomaly detection models and assessment plans [1], [2], [11], [12].

The shift toward edge-based detection of anomalies is another significant trend. As TinyML expands, there is a growing desire to do inference on embedded devices as opposed to wholly offloading to external computation [6], [7]. Such a shift applies especially to medical wearables since it can decrease response time, diminish communication overhead, and enhance privacy protection. Embedded AI thus is becoming a pertinent trend in real-time healthcare deployment, particularly in systems to be used in nonclinical settings [6]– [8], [10].

1.2.2 How Existing Systems Address the Problem

Current systems deal with cardiovascular monitoring in several ways. One is cloud-based monitoring, in which wearable or portable sensors monitor physiological information, which is sent to remote servers to be stored, analyzed and generate alerts. More computationally intensive algorithms and centralized data management are possible using this model, and it has found extensive application in digital health solutions [6], [10]. Nevertheless, it is limited to reliable connectivity and remote processing facilities.

Resolving the issue, commercial wearable health devices provide continuous monitoring access to the everyday user. They can be configured to record the heart rate and sometimes even give alerts in cases of irregular patterns being suspected. The practical value of this was that they contributed to the popularization of wearable monitoring and demonstrated that constant physiological observation could be built into the normal daily life [3], [5]. There is however a wide range in their diagnostic scope and transparency, as well as the fact that most are focused on general wellness and not specifically on elderly care.

Research prototypes have become more focused, integrating sensing, signal processing and machine learning to detect abnormalities of the heart. Experiments on benchmark data sets and AI processes indicate that automated systems can identify cardiac patterns of greater complexity [1], [4], [11], [12]. Embedded deployment has also been discussed more recently, demonstrating that it is possible to integrate lightweight intelligence with small hardware platforms that can be used in wearable systems [6]-[8].

1.2.4 Existing System Limitations

Even though there is significant improvement, there are still a few limitations in the existing systems. One such problem is cloud dependency. The system will be at the mercy of network connectivity and quality of communication when physiological data needs to be sent to external servers to be analyzed. This may cause latency, which is undesirable in cases where real-time anomaly detection is needed [6], [7]. In a home environment, the latency in detecting or raising an alert can decrease the usefulness of the monitoring solution to elderly users.

The issues of privacy and security are also significant. Physiological data is sensitive regarding health, and consistent delivery to cloud systems exposes physiological data to data leakage, unauthorized access, or data misuse. Even though cloud systems can have advantages in analytics, they may cause hesitation among the users and caregivers when they are not assured of high levels of privacy [6], [7]. This renders the issue of local processing that protects privacy a promising field of enhancement.

The other constraint is the computational cost of most sophisticated machine learning models. Developing deep learning approaches might be strong in performance, yet they are typically structured in settings with significantly more computational capabilities than are provided to wearable embedded systems [6], [8]. This introduces a disjunction between the model

performance at laboratories and the real-time performance of healthcare wearables with low power.

Lastly, there are numerous existing solutions that are not tailored to be lightweight real-time systems that can be used by the elderly. Other systems focus on broad wellness monitoring but not on clinically significant anomaly detection, and some are not designed to be cost-effective, have low power consumption, and be useful in long-term monitoring at home [3], [5], [10]. Consequently, a cardiovascular anomaly detection system that is precise, privacy-oriented, lightweight, and capable of lifelong care of elderly individuals in the real-world setting is still needed.

1.3 Research Gap

1.3.1 Limited Elderly-Focused Cardiovascular Monitoring

Despite impressive advances in wearable cardiovascular monitoring, most of the published research and a good deal of commercial systems have focused on wellness monitoring of general adult users, but not the physiological and practical application to the elderly user. The current wearables usually focus on generalized heart-rate tracking, irregular rhythm screening, or general remote monitoring and are not always tailored to the elderly-specific conditions like intermittent silent arrhythmias, in-home usage, simplified alerts, and their integration into caregiver-oriented monitoring systems [3], [5], [10]. This opens a disparity between general practice wearable monitoring and context-sensitive cardiovascular protection for elderly that is more susceptible to late detection and adverse prognosis.

1.3.2 Lack of Privacy-Preserving Edge-Based Detection

The second gap is in the reliance of most intelligent monitoring solutions on analysis with the help of clouds. Complex analytics can be supported by cloud-based architecture; however, it also brings about communication delay, connectivity reliance, and further disclosure of sensitive physiological information. The latter are of particular significance in cardiovascular monitoring since anomaly detection might require speed and since heart-related bio signals are sensitive health data. In the literature of TinyML, the concept of on-device intelligence is

increasingly viewed as a solution to the processing of data on the device and minimizing transmission to external services, but it is still not widely used in real-life cardiovascular anomaly detection systems in elderly care [6], [7]. Hence, a privacy-sensitive edge-based algorithm capable of cardiovascular analysis of the wearable device is still needed.

1.3.3 Limited Deployment of Lightweight AI Models on ESP32

There is promising effective use of deep learning in arrhythmia classification, but the effective implementation on microcontrollers is not an easy task yet. Embedded platforms like ESP32 are cheap, portable, use less power, but are also highly constrained in terms of memory, computation and runtime performance. Earlier neural network TensorFlow Lite Micro/memory-aware inference efforts demonstrated that model compression, quantization, and effective implementation plans are required to execute neural networks on this hardware [6], [8]. Nevertheless, there is still a relative scarcity of literature demonstrating relatively little translation of arrhythmia-specific deep learning pipelines into viable ESP32-based wearable systems capable of running continuously in real time. This shows that there is an evident technical mismatch between model development in research settings and deployment in constrained elderly healthcare wearables.

1.3.4 Gaps in Real-Time PPG-Based Arrhythmia Detection

PPG is also an appealing wearable cardiovascular sensing technique due to its non-invasive nature and suitability to small devices. Previous reviews have demonstrated that PPG may be used to assist in the detection of atrial fibrillation and rhythm irregularities when used in conjunction with signal processing and machine learning [3], [4]. Nevertheless, PPG-based arrhythmia detection in real-time remains a problem, and some issues remain to be solved, such as motion artifact, variable signal quality in the wrist-based application, and the inability to preserve the high detection rates with compact implemented models. The research on arrhythmia basics has traditionally been based on ECG records like MIT-BIH due to its diagnostic power [1], [2], [12], whereas PPG-based wearable detection is a more challenging implementation issue in practice. Consequently, it has remained unproven to bridge the gap to

create a low cost, real-time, PPG-driven anomaly detection system that is lightweight enough to be implemented on-device and reliable enough to be used to monitor the elderly.

1.3.5 System Comparison

The outstanding differences between current types of solutions and the suggested system are summarized in Table 1. The available wearable and cloud-based solutions have provided convenient cardiovascular monitoring capabilities, but these are not yet the full capabilities demanded by the joint needs of elderly friendliness, local privacy preserving inference, lightweight embedded intelligence and real-time anomaly detection on a wearable microcontroller-class platform [3]-[8], [10].

Table 1 – Comparison of Existing Solutions and Proposed System

Aspect	**Commercial Wearable Devices**	**Cloud-Based Monitoring Systems**	**Research AI Prototypes**	**Proposed System**
Primary purpose	General wellness / rhythm notification	Remote analytics and monitoring	Model-level arrhythmia research	Elderly-focused real-time anomaly detection
Main sensing approach	Mostly PPG / mixed wearable sensing	Wearable or clinical sensor streaming	ECG- or PPG-based experimentation	PPG-based wearable sensing
Processing location	Device + vendor platform	Mostly cloud/server side	Usually workstation / research environment	On device on ESP32
Privacy level	Moderate; data often synced externally	Lower due to server transmission	Depends on setup	High; local inference prioritized

Real-time response	Partial, platform dependent	Can be delayed by network and server processing	Often not fully embedded	Designed for direct local response
Suitability for constrained embedded deployment	Limited transparency	Low	Often not deployment-ready	Core design objective
Elderly-specific design focus	Limited	Limited	Usually, generic	Explicitly targeted
Integration with multi-sensor elderly monitoring	Rare	Possible but infrastructure-heavy	Usually, isolated module studies	Built for team-based multi-sensor integration

1.4 Research Problem

1.4.1 Main Research Problem

Although wearable sensing has advanced, PPG-based monitoring, and machine learning to analyse arrhythmia, no single solution has been established that offers low cost, preserves confidentiality, works in real time, and offers intelligence on the device to detect cardiovascular anomalies in elderly individuals. Current systems have either a high dependence on cloud connectivity, focus on overall consumer wellness, or are too computationally expensive to be feasibly implemented on the small wearable microcontrollers. Based on this, the general research issue to be discussed in this paper is that there has been no low-cost, privacy-conscious, real-time cardiovascular anomaly detection system in ageing people with on-device intelligence [3], [8], [10].

1.4.2 Research Questions

To address the above problem, this study is guided by the following research questions:

1. **How can PPG signals be used to detect cardiovascular anomalies accurately?**
   This question focuses on whether wrist-acquired PPG signals, when properly filtered and processed, can support dependable identification of abnormal cardiovascular patterns for elderly monitoring [3], [4].
2. **How can a lightweight 1D-CNN be optimized for embedded deployment on ESP32?**
   This question addresses model design, quantization, and memory-aware inference so that anomaly detection can be executed within the resource limits of microcontroller hardware [6], [8].
3. **How can real-time detection be achieved without cloud dependency?**
   This question investigates how edge inference frameworks and efficient embedded processing can enable immediate local analysis and alert generation while avoiding reliance on remote servers [6], [7].
4. **How can the solution integrate with the team's multi-sensor elderly monitoring system?**
   This question examines how the cardiovascular module can provide meaningful anomaly outputs that work alongside the project's other physiological and safety monitoring components within a broader wearable elderly care architecture [5], [10].

2. OBJECTIVES

2.1 Main Objective

- To design and develop a privacy-focused real-time cardiovascular anomaly detection system for elderly individuals using Edge AI

2.2 Specific Objectives

- To collect real-time PPG signals using the MAX30102 sensor
- To preprocess and filter the acquired cardiovascular signals
- To develop and train a 1D-CNN model for arrhythmia detection
- To optimize and deploy the trained model on the ESP32 microcontroller
- To evaluate the performance of the system using accuracy and real-time response metrics
- To integrate the cardiovascular module with the team's overall wearable health monitoring system

3. METHODOLOGY

3.1 Research Approach

This proposal will take the design and development research approach to develop a wearable cardiovascular anomaly detector module in the elderly population. The study is solution based and aims at the design, implementation and test of an effective embedded system capable of performing in real time. Furthermore, to test the performance of the proposed model, experimental validation method is applied to both the condition of prepared data sets and prototype-level deployment. The study thus comprises algorithm development, embedded implementation and performance testing in a prototype-based research approach [1], [4], [6].

3.2 Technologies, Techniques, and Tools

3.2.1 Hardware Components

The hardware platform is comprised of an ESP32 microcontroller, a MAX30102 PPG sensor, and battery-powered auxiliary wearables. The ESP32 is chosen due to its low cost, utility power, and adequate embedded processing to perform lightweight on-device inferences. Photoplethysmography signals are obtained by means of the MAX30102 due to the non-invasive nature of PPG sensing and its appropriateness in wearable cardiovascular monitoring [3], [6]. Additional components that support the wearable, which includes rechargeable battery and simple packaging material are provided to make the prototype portable.

3.2.2 Software Tools

Software environment contains Python to write a model, TensorFlow to train a neural network, TensorFlow Lite to Microcontrollers to perform inference on an embedded system, and Platform IO with Visual Studio Code to deploy an ESP32. The preparation and cleaning of the physiological data prior to inputting them into the model is performed with signal preprocessing libraries. The tools aid the entire process of dataset preparation to microcontroller implementation [6], [8].

3.2.3 Techniques and Algorithms

The key technical processes include PPG signal capture, noise reduction and preprocessing, 1D Convolutional Neural Network classification, and TinyML implementation. The sensing mode adopted is PPG since earlier research indicates that it can be used in cardiovascular monitoring and rhythm analysis of the wearable [3], [4]. The reason behind the adoption of a 1D-CNN is that convolutional models are successful in discovering temporal trends in biomedical signals and TinyML allows the implementation of models on embedded systems with limited computational capacity [4], [6].

3.3 High-Level System Architecture

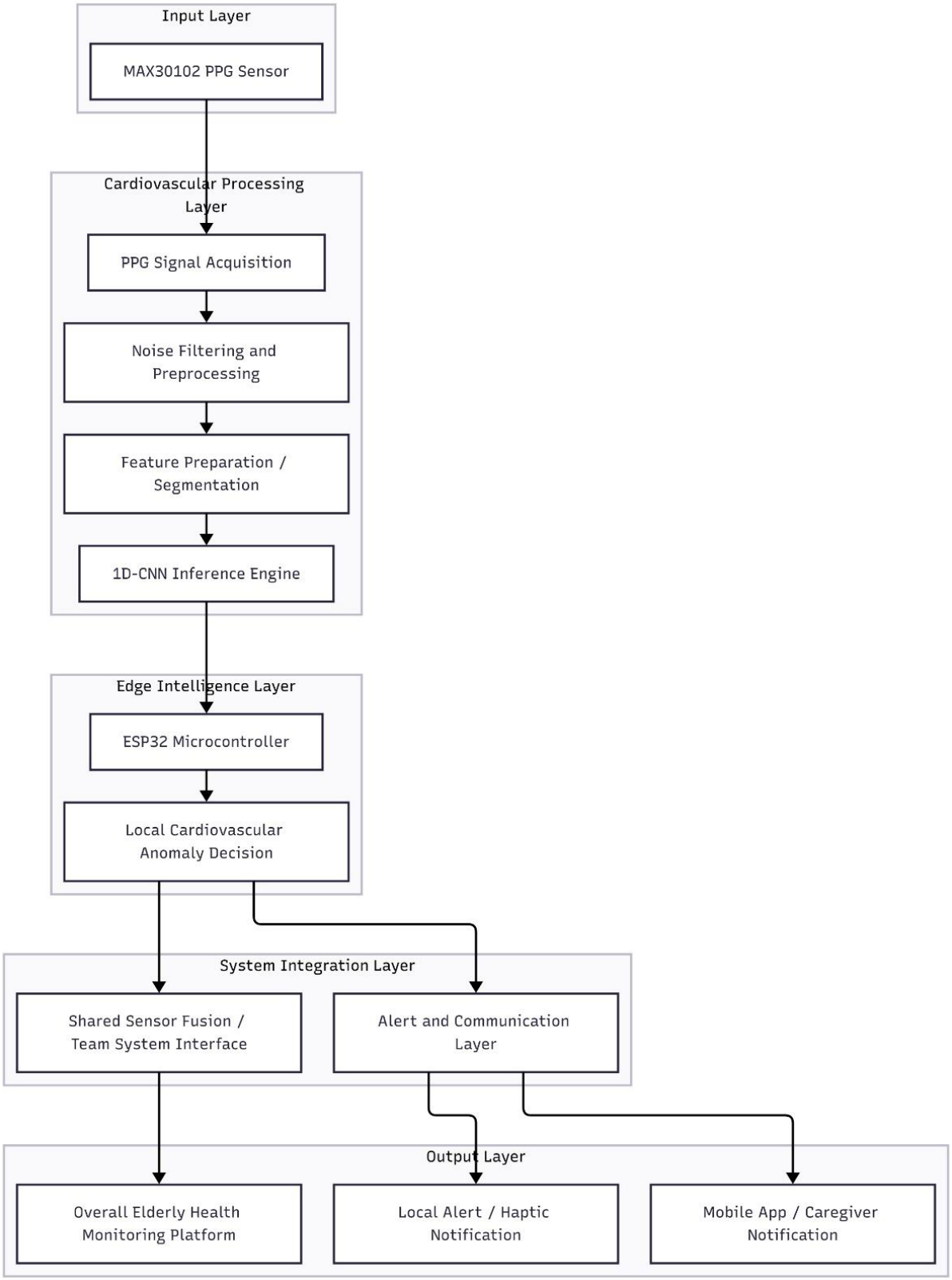


Diagram 1- High level Architecture diagram

The cardiovascular module in the general team architecture will serve as the heart-monitoring unit of the multi-sensor elderly care system. It is meant to receive cardiovascular data, conduct local anomaly analysis and give formal outputs to the common alert and communication layer. The data flow is initiating at the PPG sensor then going through edge processor to perform preprocessing and classification and finally to the anomaly output layer where the output is utilized to give a local warning or integrate with the full wearable platform.

3.4 Proposed System Overview

The suggested system adheres to the end-of-end workflow that is initiated by sensor-based data collection and terminated by the production of anomaly alerts. To start with, the sensor of the MAX30102 records PPG on the user. The signal obtained is further preprocessed locally to minimize noise and enhance the quality of the signal. These clean signal images are then sent to a 1D-CNN model (lightweight) that classifies them. According to the model output, ESP32 decides whether the cardiovascular pattern is abnormal or normal. In case of an abnormal pattern, the system sends out an alert, which can be sent over to the broader team-based monitoring platform.

3.5 Component Process Flow

The proposed component has five consecutive steps that are part of its process flow: PPG sensor input, signal preprocessing, feature preparation, 1D-CNN classification, and anomaly output. Wearable PPG signals are prone to motion artefacts and noise, hence signal preprocessing is required [3], [4]. Once the signal has been filtered and segmented, it is then passed into the trained model where it is classified. The ultimate output will determine whether the cardiovascular pattern that was observed is normal or abnormal.

3.6 Dataset and Model Development

3.6.1 Dataset

As is the primary reference dataset to develop and validate the model, the MIT-BIH Arrhythmia Database is utilized. It is an established standard of arrhythmia studies and has been broadly applied to assess automated cardiovascular classification algorithms [1], [2], [12].

3.6.2 Model Development

Approximation features in model development encompass data preparation, training, validation, optimization and quantization. The preferred data undergo cleaning, segmentation and supervised learning formatting. The cardiovascular patterns are then trained on the 1D-CNN. Following training, the model is optimized to the best to minimize memory and computation needs. Quantization is used to enable embedded implementation via TensorFlow Lite for Microcontrollers [6], [8].

3.7 Implementation Procedure

The start is with hardware installation and sensorization on ESP32 platform. Then signal acquisition and preprocessing routines are written. The trained model is compiled into a TinyML-native code and loaded onto the microcontroller. Lastly, there is real-time testing, which is done to test the prototype behaviour in live sensing conditions.

3.8 Performance Evaluation

3.8.1 Evaluation Metrics

Classification performance is measured by accuracy, precision, recall and F1-score which are evaluated to assess the proposed system. Besides that, inference time and memory consumption are also considered to determine embedded feasibility, as microcontroller-based healthcare monitoring requires lightweight execution [6], [8].

3.8.2 Validation Method

There are two stages of validation. First, prepared dataset samples are used to test the trained model and study the classification effectiveness. Second, testing real-time sensor input on the prototype is carried out to observe embedded functionality in real-life conditions. The generated outputs are compared with the anticipated anomaly labels and system behaviour requirements to evaluate the soundness of the suggested module [1], [4], [6].

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

The functional requirements focus on the main activities of the cardiovascular monitoring module, beginning with physiological signal acquisition and ending with communication of detection results to the integrated team system.

Table 2 – Functional Requirements

Requirement ID	Functional Requirement	Description
FR1	PPG Data Acquisition	The system shall acquire photoplethysmography data from the MAX30102 sensor for continuous cardiovascular monitoring.
FR2	Real-Time Signal Preprocessing	The system shall preprocess cardiovascular signals in real time to reduce noise and improve signal quality before analysis.
FR3	Rhythm Classification	The system shall classify the detected heart rhythm as normal or anomalous using the embedded 1D-CNN model.
FR4	Alert Generation	The system will generate an alert when an abnormal cardiovascular condition is detected.

FR5	System Integration Output	The system shall transmit classification results to the integrated team platform for further monitoring and coordinated response.

4.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes that are necessary for practical deployment. Since the proposed solution is intended for embedded elderly healthcare monitoring, these requirements are especially important for ensuring dependable and efficient system operation.

Table 3– Non-Functional Requirements

Requirement ID	Non-Functional Requirement	Description
NFR1	Real-Time Performance	The system should process input data and generate anomaly detection results with minimal delay.
NFR2	Low Power Consumption	The system should operate efficiently on battery power to support wearable continuous monitoring.
NFR3	Privacy Preservation	The system should minimize external data transmission by performing analysis locally on the ESP32 device [6], [7].
NFR4	Reliability and Accuracy	The system should provide consistent and accurate cardiovascular classification under expected usage conditions [4], [5].
NFR5	Scalability and Maintainability	The module should support future updates, model improvements, and integration with additional sensing components.
NFR6	User Safety and Usability	The system should remain safe, comfortable, and easy to use for elderly individuals during continuous wearable operation.

4.3 Hardware and Software Requirements

The proposed cardiovascular anomaly detection module requires both hardware and software resources to support wearable data acquisition, embedded inference, and system integration. The hardware components are selected to ensure portability, low power consumption, and compatibility with real-time physiological monitoring. The software tools are chosen to support signal preprocessing, model development, microcontroller deployment, and embedded testing. Since the proposed system is based on local processing, the selected tools must also support efficient implementation on constrained edge hardware [3], [6], [8].

Table 4– Hardware Requirements

Item	Description	Purpose
ESP32	Low-cost microcontroller with embedded processing capability	Executes local preprocessing, model inference, and communication functions
MAX30102	PPG-based heart rate and pulse oximetry sensor	Captures cardiovascular-related PPG signals for anomaly detection
Battery Module	Portable power supply unit	Enables wearable and continuous operation of the prototype
Wearable Housing Components	Casing, straps, connectors, and mounting materials	Supports physical integration and user-friendly wearable deployment

Table 5– Software Requirements

Software Tool	Description	Purpose
Python	Programming environment for data handling and model development	Used for preprocessing, training, and evaluation
TensorFlow	Machine learning framework	Develops and trains the 1D-CNN classification model

TensorFlow Lite Micro	Embedded inference framework	Supports TinyML deployment on the ESP32
PlatformIO	Embedded development platform	Uploads, configures, and tests firmware on the microcontroller
Data Preprocessing Tools	Signal processing libraries and utilities	Clean, segment, and prepare physiological data for model input

4.4 Expected Test Cases

The test cases are designed to verify sensing accuracy, preprocessing performance, model behavior, alert generation, and module integration with the overall team system.

Table 6 – Expected Test Cases

Test Case ID	Test Scenario	Expected Outcome
TC01	Sensor reads valid PPG signal	The MAX30102 captures a stable and usable PPG waveform
TC02	Noise filtering works correctly	Preprocessed signal shows reduced noise and improved clarity
TC03	Model detects normal rhythm correctly	The system classifies normal cardiovascular patterns accurately
TC04	Model detects anomaly correctly	The system identifies abnormal cardiovascular patterns correctly
TC05	Alert is triggered on abnormal detection	A warning signal is generated when an anomaly is detected
TC06	Module communicates output to full system correctly	Classified output is successfully transferred to the team integration layer

5. DESCRIPTION OF PERSONNEL AND FACILITIES

5.1 Personnel

The proposed component is undertaken by a single student researcher who is responsible for the design and development of the cardiovascular anomaly detection module. The assigned role includes both technical and documentation-related tasks required for successful completion of the individual research contribution.

Table 6 – Personnel Details

Item	**Details**
Student Name	Abeysinghe S.A.M.M.
Student ID	IT22081148
Assigned Role	Development of Cardiovascular Anomaly Detection Module

Table 7 – Responsibilities of the Student Researcher

Responsibility Area	**Description**
Literature Review	Review scholarly studies related to wearable cardiovascular monitoring, arrhythmia detection, and Edge AI-based embedded systems
Sensor Integration	Configure and integrate the MAX30102 sensor with the ESP32 platform for physiological signal acquisition
Model Development	Develop, train, optimize, and prepare the 1D-CNN model for embedded deployment
Testing and Evaluation	Conduct dataset-based validation and real-time prototype testing using selected evaluation metrics
Documentation	Prepare technical diagrams, report content, implementation details, and evaluation findings

5.2 Facilities and Resources

The project will be supported by a combination of computing resources, software tools, datasets, and embedded hardware components necessary for prototype development and validation.

Table 8 – Facilities and Resources

Resource	**Purpose**
Personal Computer / Development Machine	Used for model training, preprocessing, documentation, and code development
Internet Access	Supports literature access, dataset retrieval, and software package installation
VS Code / PlatformIO Environment	Used for ESP32 programming, firmware development, and debugging
Required Datasets	Provide benchmark data for model training and validation
Embedded Hardware Components	Enable prototype implementation and real-time physiological sensing
Testing Environment	Supports functional testing and prototype validation under expected operating conditions

6. COMMERCIALIZATION PLAN

The cardiovascular anomaly detection module suggested in this paper can be practically commercialized due to its response to a relevant healthcare concern in elderly monitoring, namely a wearable affordable and privacy-oriented device. The main target market is the ageing and independent elderly population, family caregivers, and assisted care facilities that need constant cardiovascular monitoring and emergency preparedness. The second target market consists of hospitals, rehabilitation facilities, or home healthcare providers who require remote monitoring help at a reasonable price. Here, the customer persona may be defined as an elderly customer, who may need constant heart monitoring, a caregiver, who may need timely

notifications in the case of an abnormal event, and a healthcare provider, who may be interested in patient monitoring solutions that can be scaled.

The proposed system has a value proposition of providing real-time detection of cardiovascular anomalies based on on-device intelligence. In contrast to the systems where operation is entirely based on cloud processing, the offered solution focuses on local processing, minimized delay, enhanced privacy protection, and convenient use in home-based geriatric care. This gives a high benefit to users who need urgent notifications and confidential management of delicate health data.

There are a variety of methods to build the revenue model. The former is direct sales of wearable devices to individual users and caregivers. The second one is a subscription-level caregiver notification service capable of offering more alert service and monitoring. The third is institutional sales to the eldercare organizations, hospitals, and home healthcare agencies that might embrace the system to monitor patients.

The cost estimation will be based on prototype production cost, approximate cost per unit of scaled production, and optional support or maintenance costs. Following this structure, the pricing approach must be affordable to individual users but have tiered packages where institutional clients can be offered. High-end features could be offered as an optional subscription fee.

Competitive advantage in the proposed solution is also provided in terms of Edge AI processing, improved privacy protection, reduced response time, and compatibility with a broader multi-sensor elderly monitoring ecosystem. Intellectually, commercialization can be facilitated by software copyright protection, ownership of the trained lightweight model, and a potential patent on the embedded healthcare monitoring workflow.

7. BUDGET AND JUSTIFICATION

The suggested cardiovascular anomaly detection module will cost a relatively small amount of money due to the design using cheap embedded hardware and mostly open-source software solutions. As the system is an edge-based architecture, no compulsory cloud spending is made

at the prototype level, which also aids in lowering the total implementation cost. The primary budget items consist of hardware, software, optional costs of service, and approximated human effort needed in development, testing, and documentation.

Table 9 – Hardware Costs

Item	Estimated Cost (LKR)	Purpose
ESP32 Microcontroller	2,500	Embedded processing, model inference, and communication
MAX30102 Sensor	1,800	Acquisition of PPG-based cardiovascular signals
Battery and Charging Components	2,000	Portable power supply for wearable operation
Wiring and Wearable Accessories	1,200	Physical integration and prototype assembly
Total Hardware Cost	**7,500**	

Table 10– Software and Service Costs

Item	Estimated Cost (LKR)	Purpose
Open-Source Tools	0	Python, TensorFlow, TensorFlow Lite Micro, and PlatformIO are used without licensing cost
Optional Licensed Tools	0 / Not Required	No paid tools are required at the current prototype stage
Optional Future Cloud Dashboard	2,500	Possible future cost if remote dashboard services are introduced
Total Current Software/Service Cost	**0**	

Table 11 – Human Effort Cost

Activity	Estimated Cost (LKR)	Purpose
Design and Development Time	8,000	Hardware setup, preprocessing logic, and system integration
Model Training and Testing Time	6,000	Model preparation, validation, and embedded testing
Documentation Effort	4,000	Proposal writing, technical documentation, and reporting
Total Human Effort Cost	**18,000**	

8. WORK BREAKDOWN STRUCTURE (WBS)

Work Breakdown Structure will identify the large tasks that this single component will perform and how the cardiovascular anomaly detection module fits within the larger team project. As this proposal is a specialized subsystem of the entire elderly health monitoring solution, the task structure has both independent tasks and teamwork. The personal activities are dedicated to the research, design, development, testing, and documentation of the cardiovascular module, whereas the team activities assure the correspondence to the common architecture of the system, communication system, and the eventual integrated validation.

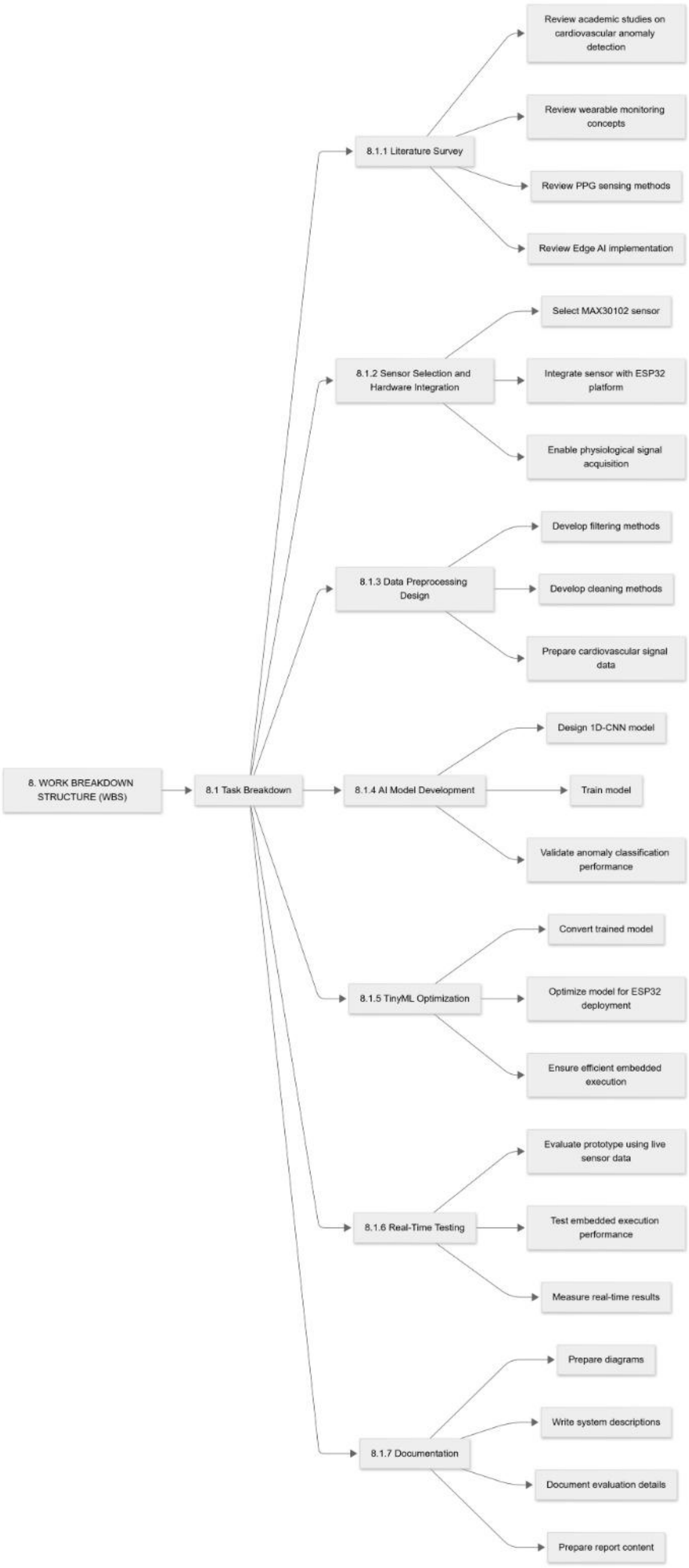
Table 12 – Abeysinghe's Main Tasks

WBS ID	Task	Description
8.1.1	Literature Survey	Review academic studies on cardiovascular anomaly detection, wearable monitoring, PPG sensing, and Edge AI implementation
8.1.2	Sensor Selection and Hardware Integration	Select the MAX30102 sensor and integrate it with the ESP32 platform for physiological signal acquisition
8.1.3	Data Preprocessing Design	Develop filtering, cleaning, and signal preparation methods for cardiovascular data
8.1.4	AI Model Development	Design, train, and validate the 1D-CNN model for anomaly classification
8.1.5	TinyML Optimization	Convert and optimize the trained model for efficient deployment on the ESP32 microcontroller
8.1.6	Real-Time Testing	Evaluate prototype performance using live sensor data and embedded execution
8.1.7	Documentation	Prepare diagrams, system descriptions, evaluation details, and written report content

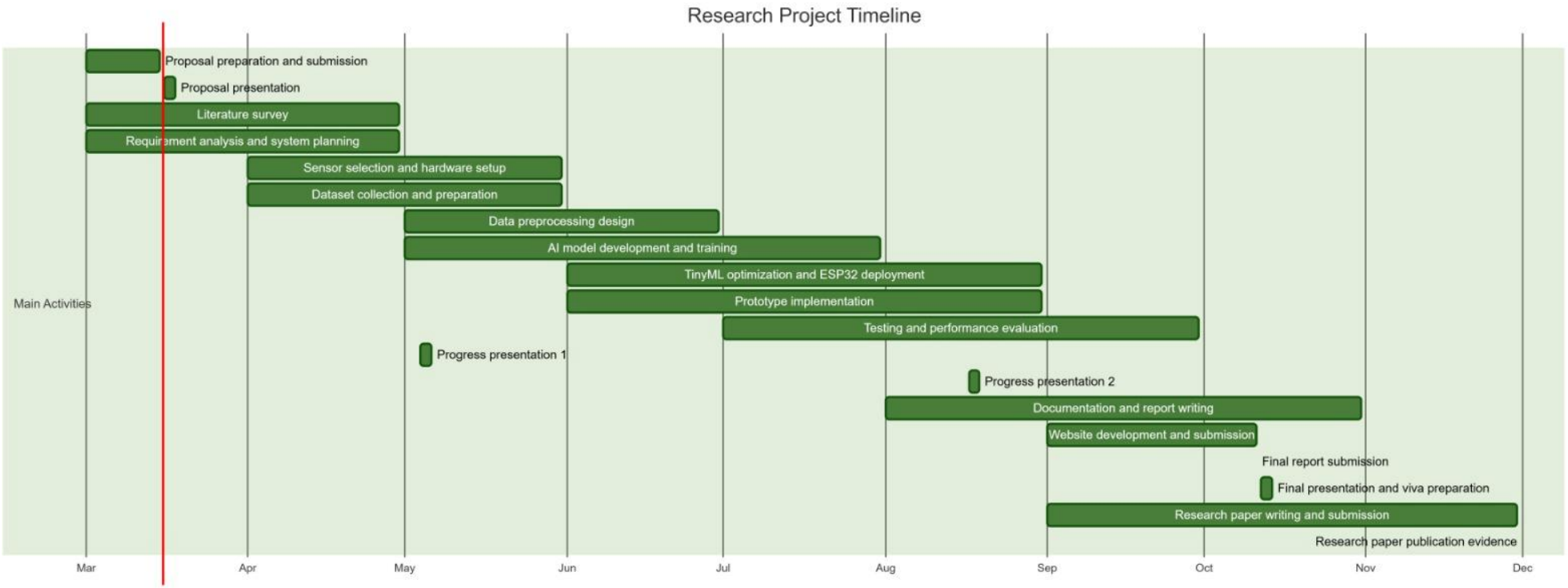
Table 12 – Team Collaboration Tasks

WBS ID	Collaborative Task	Description
8.2.1	Integration with Full Team Architecture	Connect the cardiovascular module with the broader multi-sensor elderly monitoring system
8.2.2	Shared Alert and Communication Layer	Support the common output and notification mechanism used across all project components
8.2.3	Shared System Validation	Participate in integrated testing to confirm compatibility, coordinated operation, and end-to-end system functionality

Diagram 2- Work breakdown structure



9. GANTT CHART



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